

QuForge Industrial Studio: Professional User Manual

This manual is designed to take a beginner from their first launch to advanced programmatic quantum circuit design. The **QuForge Industrial Studio** is a 10-qubit Electronic Design Automation (EDA) tool that simulates quantum hardware physics, including noise, decoherence, and connectivity constraints.

1. Introduction

QuForge is an integrated environment for designing, simulating, and optimizing quantum algorithms.¹ Unlike standard simulators, QuForge focuses on **Hardware-Aware Design**, meaning it accounts for the physical limitations of a 10-qubit processor ring, such as T_1 relaxation and electromagnetic interference (EMI).

2. Interface Breakdown

A. The Design Canvas (Visual EDA)

- **The Grid:** Each horizontal line represents a **Qubit** (Q_0 to Q_9). The vertical dashed lines represent time in nanoseconds (ns).
- **Zooming:** Use **Ctrl + Mouse Wheel** to zoom in for precise gate placement or zoom out to see the entire algorithm timeline.
- **Scrolling:** Use the horizontal scrollbar at the bottom to navigate longer circuits.

B. Calibration Sidebar

This panel allows you to simulate different hardware environments:

- **T_1 (Relaxation):** The time it takes for a qubit to lose its energy.² Higher values mean more stable qubits.
- **T_2 (Dephasing):** The time it takes for a qubit to lose its quantum phase. This is critical for interference patterns.
- **Gate/Readout Error (%):** Adjust the statistical probability of a hardware failure during an operation or measurement.

C. Industrial Resource Report

- **Live Feedback:** Displays the calculated **Fidelity** of your circuit (how "pure" the final state is) based on your calibration settings.
- **State Vectors:** Shows the top 8 most probable quantum states and their complex amplitudes.

3. Feature Guide

Interactive Tools

- **Crosstalk Map:** Generates a heatmap showing which qubits are suffering from high noise levels based on simultaneous gate operations.

- **Topology Map:** Visualizes the physical structure of the 10-qubit processor. It highlights active qubits and the links being used by your algorithm.
- **EMI Error Button:** A specialized diagnostic tool. If gates are placed too close together vertically, a "**Collision**" error is triggered, indicating pulse interference.

Advanced Visualizers

- **3D Vector Analysis:** Select a qubit to see its state represented on a **3D Bloch Sphere**.
- **Pulse Timing View:** Shows the actual microwave pulse signals sent to the hardware. Essential for timing-sensitive research.
- **Coherence Map:** Displays the density matrix to visualize the level of entanglement between all 10 qubits.

4. Step-by-Step Guidance

How to Build a Circuit

1. **Select a Gate:** Click on a gate icon (**H**, **X**, **Y**, **Z**, or **M** for Measurement) from the top bar.
2. **Place Gate:** Click on the qubit line at the desired time.
3. **CNOT (Entanglement):** Click the CNOT button, click the **Control Qubit**, then click the **Target Qubit**.
4. **Delete:** Right-click any gate on the canvas to remove it.

How to Use the Python Editor

1. **Open Editor:** Click the **PYTHON SCRIPT** button.
2. **Coding:** Use the `self.sim` object to write custom scripts. This is perfect for loops and complex algorithms like VQE.
3. **File Management:**
 - Use **OPEN** to load a .py script.
 - Use **SAVE** or **SAVE AS** to store your algorithm for later.
4. **Execute:** Click **RUN ALGORITHM** to see the script generate gates on the visual canvas automatically.

5. Efficiency Tips for Beginners

- **Stagger Your Gates:** To avoid EMI errors and improve fidelity, avoid placing gates on different qubits at the exact same time (ns). Shift them slightly horizontally.
- **Batch Run:** Use the **BATCH RUN** button at the bottom to simulate **1024 "shots"** of your circuit to get a statistical probability distribution.
- **Transpile Often:** Use the **TRANSPILE** button to clean up your circuit. It automatically removes redundant gates (like two 'X' gates in a row) to save on hardware error.